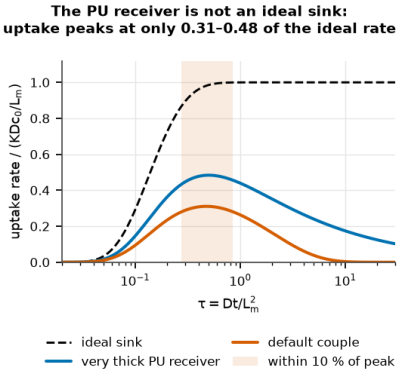
Computational design toolkit for a polymer electrolyte experiment (Li–S batteries)

Kaiming Zhang

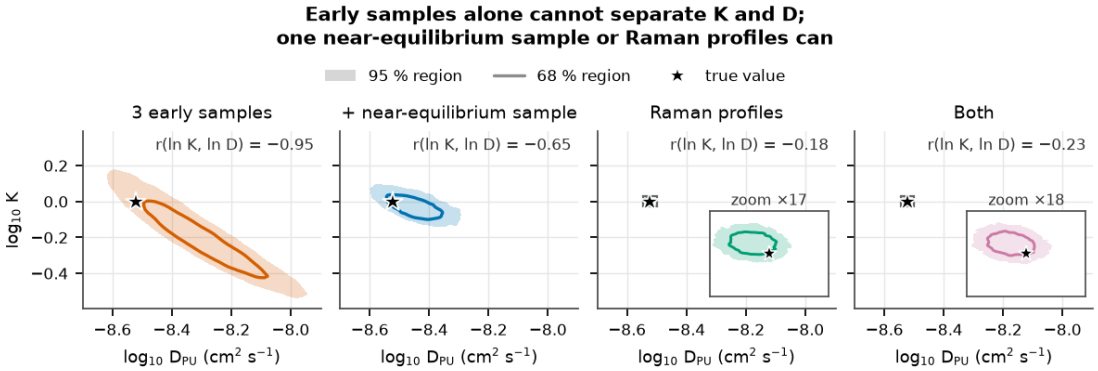
Motivation. Seven polyurethane (PU) solid electrolytes, sharing one HDI/1,4-butanediol hard segment and differing in one soft-segment variable at a time (four polyethers, two polycarbonates, one polyester), are planned to separate the solubility, mobility and chemical consumption of lithium polysulfides. Before any polymer exists, I used computation to design the transport experiment, test the analysis and fix testable predictions.

Diffusion-couple model: designing the transport experiment

Problem: which sampling times and geometry make the partition coefficient K and diffusivity D measurable in the planned PEO-donor | PU-membrane | PU-receiver couple? *Method:* a 1-D finite-volume model verified against an exact Laplace-domain solution; Fisher-information design, Bayesian inversion (emcee) and global sensitivity analysis. *What it changed:* full-model fits instead of the slope method; the first sample (0.1 L^2/D, below quantification in 22 of 28 scenarios) moved later; a near-equilibrium sample or Raman profiles added so that K and D separate; the UV-vis detection limit measured first.



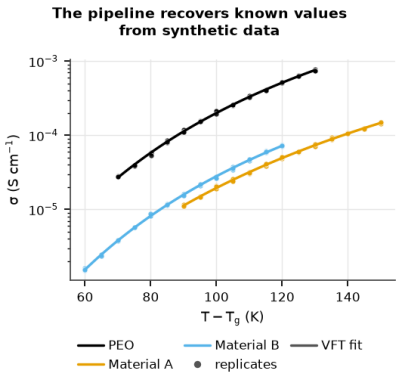
What you see: receiver uptake rate relative to the ideal steady rate K D c_0/L_m (assumed inputs): the PU receivers peak at only 0.31–0.48 of it (shaded: default couple within 10 % of its peak, τ = 0.27–0.84). **What it means for the experiment:** the time-lag slope would underestimate the permeability, so K and D are fitted with the full model.



What you see: posterior regions of K and D_{PU} from synthetic data (assumed $D_{PU} = 3 \times 10^{-9}$ cm^2 s^{-1}, K = 1). Three early samples give a long ridge (r = −0.95); a near-equilibrium sample or Raman profiles shrink it (r = −0.65, −0.18, −0.23). **What it means for the experiment:** the plan includes one near-equilibrium sample or Raman profiles.

Analysis pipeline with uncertainties

Problem: impedance, polarisation, DSC, UV-vis, FTIR, diffusion-couple and cell data must become numbers with uncertainties. *Method:* an automated CSV-based pipeline (circuit fits, VFT, Bruce–Vincent F^+, saturation limit, couple inversion) with Monte Carlo errors. *What it changed:* tests on synthetic data with known truths made the impedance fits modulus-weighted and the saturation limit subject to a significance test, and showed that low-R Li_2S_8 films need LiTFSI reference films at the same R.

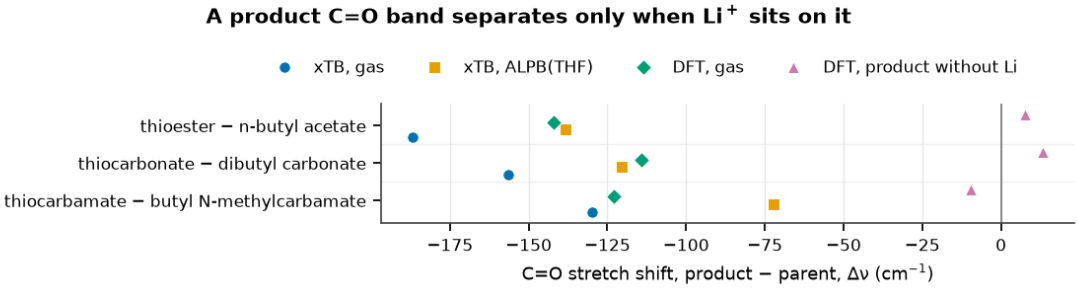


What you see: synthetic test data (PEO, materials A and B): σ against T − T_g with automated VFT fits; 27 of 37 known quantities were recovered within 3 %. **What it means for the experiment:** the analysis is tested before the first real sample.

Full code available on request.

Molecular calculations: xTB screening with DFT checks

Problem: do model molecules of the soft segments bind Li^+ and polysulfides differently enough to matter, and where would reaction products appear in FTIR? *Method:* GFN2-xTB with many starting structures, counterpoise-corrected B3LYP-D3(BJ) checks and DFT harmonic frequencies. *What it changed:* binding differences of a few kJ mol^{-1} are not used, and the FTIR test gets an un-aged control and NMR of model compounds.

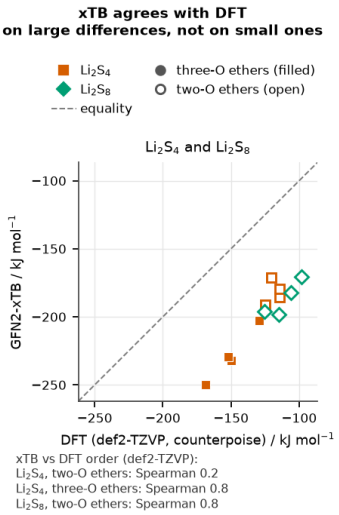


What you see: computed C=O shift of each thio-product against its parent C=O (GFN2-xTB and B3LYP-D3(BJ)/def2-SVP): 114–142 cm^{-1} lower when the Li of S_4Li sits on the product C=O, within 14 cm^{-1} without Li (DFT). **What it means for the experiment:** FTIR can miss the reaction, so an un-aged control and model-compound NMR are added.

Predictions written down before the experiment

Problem: can the calculations predict the experiment? *Method:* before the first synthesis I fixed 43 rule-based entries (22 with a predicted direction), each with a threshold or interval, a decision rule (supported / not supported / inconclusive against the combined 1σ uncertainty) and a scoring script. *What it changed:* in version 2, still before any measurement, three-oxygen ether models became primary, and every polysulfide-based direction must also hold in counterpoise-corrected DFT; 7 xTB directions lacked this confirmation.

What you see: xTB against counterpoise-corrected DFT (def2-TZVP) for the Li_2S_4 and Li_2S_8 complexes: the order agrees for the sets with large differences (Spearman 0.8 and 0.8), not for Li_2S_4 with two-O ethers, which differ by only 11 kJ mol^{-1} (Spearman 0.2). **What it means for the experiment:** only above-threshold xTB differences confirmed by DFT become predictions.



Methods & tools: finite-volume model with Laplace-domain verification; emcee Bayesian inversion; global sensitivity analysis; GFN2-xTB; B3LYP-D3(BJ) with counterpoise correction; harmonic frequencies; pytest (61 tests), seeded runs.

Limitations: all inputs are assumptions and there are no experimental data yet; single model molecules cannot represent the polymer; DFT was used at validation level only. The code was written with the assistance of the AI coding assistant Claude Code (Claude Opus 5.5, max effort); the scientific questions, mappings and decision rules were set by me.

Full code available on request.